Rank-27 Decision, Congruence, and Substitute Consolidation

Tactix Development Project

26 August 2026

Status

The combined source, complete checked Tier-B plus predeclared Tier-C gate, and mandatory Claude-owned exact-source sanitizer fuzz campaign are terminal PASS. The source is ready for promotion as one isolated increment; no next solver repair has been mixed into it.

The candidate is SHA-256 `acf9e716e78ea8bded71ee5121a975223f3510fcc72f84fab79865903aa2e013` on top of accepted solver-source commit `d8da6a17dc536b4a1c399d487582bcca48476b35`; later repository commits before this promotion contain documentation and fuzz-resource tooling only. Its three general implementation changes remove avoidable diagnostic and maintenance work without introducing benchmark identity, rank, deadline, or per-instance policy.

1 Changes and boundaries

Decision journal. Production builds no longer construct and shift the dynamic `literal=` candidate string for every decision when the rule journal is compiled out. Journal-enabled builds construct the same string lazily and retain exact record/replay identity. The decision transaction, branching policy, heap state, scheduling, and proof behavior are unchanged.

Congruence lifecycle. Forward subsumption reconnects the live large-watch image and defers physical arena reclamation to the caller's established phase boundary. This removes a duplicate full-arena pack immediately before Substitute. The live clause order is unchanged, stable references remain usable, no watch points into garbage, and the later boundary still owns forwarding and arena-generation invalidation. A strengthened standalone test checks both pre-boundary and post-collection states.

Substitute affected owners. Canonical Substitute marks every retiring literal and every binary mirror owner incident to it. It retains the former global import/sign order and per-list order over precisely that induced population, avoiding global scans and compactions of unrelated watch lists. Sweep/session behavior, SCC work, proof work, candidate order, large-clause rewriting, and collection remain unchanged.

These remain representation and lifecycle repairs below the proof-rule/ strategy boundary. N1–N4, exact logical tick/cost, candidate order, scheduling order, learning, pruning, and retained semantic state remain the acceptance surface.

2 Exact-state evidence

The decision repair preserves the accepted 64-row event trace SHA-256 `7a6216bb...7cc1` and all 175,272,633 Rank-27 decisions while lowering internal time 365.249 to 359.799 seconds. Its predeclared

B+C screen passes 12/12 checked results and lowers paired total 635.585 to 632.261 seconds.

The Congruence repair removes exactly two productive prefix collections. Its full Rank-27 pair is exact at normalized SHA-256 `10372d94...f1ffd0`; internal time is 358.115 to 356.788 seconds and Congruence 14.432 to 13.366 seconds. The six-instance checked B+C screen passes 12/12; Congruence improves on all six and sums 43.935 to 42.004 seconds. Total B+C time is noisy-negative by 0.52%, so retention relies on the exact Rank-27 owner/total win and the demonstrated duplicate lifecycle, not the short aggregate alone.

The Substitute repair preserves the same full Rank-27 normalized hash and lowers internal time 358.157 to 354.539 seconds. Substitute falls 21.704 to 18.528 seconds and maximum RSS falls 672,116 to 608,084 KiB. Its checked B+C screen passes 12/12 and improves Substitute on all six instances. The two nonexact cost images differ only because omitted unrelated-list compactions no longer refresh conservative binary-first metadata; formula, watch contents and order, decisions, conflicts, propagations, rule effects, schedule, learning, pruning, and verdict remain equal.

All focused Release/N1–N4/proof/boundary/reset guards pass for each isolated increment. The final exact-source Release suite is functionally 73/74: all 73 solver, proof/model, scheduling, trace, boundary, and cleanup tests pass. Only the already recorded documentation-size budget fails.

3 Complete consolidation gate

Frozen campaign `20260826-1300-BC-decision-congruence-substitute-consolidation` owns all 28 Tier-B and five predeclared Tier-C identities, immutable matrix SHA-256 `9cc79541...c30e2`, candidate and accepted pins, and validators. Array `1648120` and aggregate `1648129` are terminal PASS.

Gate property	Candidate	Accepted baseline
Valid results	33/33	33/33
SAT models independently checked	17	17
UNSAT proofs independently verified	16	16
PAR-2 / solve total (s)	1446.060	1449.704
Peak RSS (GiB)	2.914	2.930

Candidate/baseline PAR-2 ratio is 0.997486. Missing tasks, invalidity, solve loss, regressions, cleanup failures, and certificate residue are all zero. Summary SHA-256 is `b57977a...eec4`. Every solver invocation is an independent SLURM task under the 16-GiB/1200-second contract.

4 Claude-owned fuzz gate

Claude Code's authoritative terminal report classifies campaign `20260826-1330-decision-congruence-substitute-consolidation-fuzz` PASS and updates its manifest to `terminal-pass`. Codex accepted this report without inspecting fuzz logs or artifacts.

All five generated legs complete 50,000/50,000 cases with zero failures. Claude reports 20,927 independently checked SAT models, 17,727 native-LRAT UNSAT proofs accepted by the pinned checker, and Kissat agreement on all 38,654 verdict cases. Corpus replay is 135/135 with zero failures. The dedicated reach phase terminates 40/40 oracle-clean cases and exceeds every required lazy-journal, deferred-collection, affected-owner, productive/zero-image, and session floor. There is no sanitizer finding, crash, invariant abort, rejected model/proof, or differential disagreement. Bounded five-second nonterminations carry no oracle claim; exactly 25 diagnostic inputs per leg are retained. Certificate, proof, model, checker, and node-local scratch residue are zero.

No job was cancelled or replaced. Congruence- and Sweep-stress jobs `1648215/1648217` complete their full accepted summaries before SLURM records exit 127. This is a post-oracle infrastructure artifact: commit `8a9c8f7` changed a comment length in the live `run.sh` while both Bash interpreters were blocked inside `fuzz.py`; their stale byte offsets then entered the shifted final line after `fuzz.py` had exited zero. Claude records this as neither a solver nor fuzz-acceptance failure. Future fuzz jobs must execute an immutable harness snapshot.

5 Promotion and next action

The complete B+C and fuzz gates authorize promotion, report completion, commit, and push. First close the newly exposed immutable-harness-snapshot tooling defect without changing solver behavior. Then freeze the promoted pin and immediately resume the next demonstrated current-Tactix/current-Kissat common-state gap.

The separate cluster timing study is not part of this source increment. Its approved `-c 12`/48,000-MiB, three-paired-wave decision-grade policy is already committed; fuzzing remains one campaign and is not repeated by that timing rule.